

Real-Time Color Detection and Tracking System using OpenCV

1. Introduction

Computer vision is an important field in artificial intelligence that enables machines to interpret visual data. One of the fundamental tasks in computer vision is detecting and tracking objects based on color.

This project focuses on building a real-time color detection system using Python and OpenCV. It identifies a specific color (red) from a live webcam feed and highlights it.

2. Objective

- To detect a specific color in real-time
 - To understand image processing techniques
 - To implement contour detection
 - To build a simple and practical computer vision application
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3. Problem Statement

In real life, tracking colored objects is useful in areas such as robotics, surveillance, and industrial automation. However, detecting objects accurately in real-time can be challenging due to variations in lighting and background noise.

This project addresses this problem by using HSV color space and contour detection techniques to identify and track colored objects.

4. Tools and Technologies

- Python
 - OpenCV
 - NumPy
 - Webcam
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5. Methodology

Step 1: Capture Video

The webcam captures live video frames continuously.

Step 2: Convert to HSV

Each frame is converted from BGR to HSV color space for better color detection.

Step 3: Define Color Range

Two HSV ranges are used to detect red color accurately.

Step 4: Create Mask

A mask is created to isolate only the red color from the frame.

Step 5: Detect Contours

Contours are detected from the mask to identify the object.

Step 6: Draw Bounding Box

A bounding rectangle is drawn around the detected object.

6. Output

- Live camera feed showing the detected object
 - Mask window displaying the detected color
 - Bounding box around the red object
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7. Challenges Faced

- Lighting conditions affecting detection accuracy
 - Noise in the captured image
 - Selecting the correct HSV range for red color
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8. Results

The system successfully detects red-colored objects in real-time with good accuracy under proper lighting conditions.

9. Learning Outcomes

- Learned the basics of OpenCV
- Understood HSV color space

- Implemented contour detection
 - Developed a real-time computer vision application
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10. Future Scope

- Add support for multiple colors
 - Improve accuracy using advanced techniques
 - Add object recognition features
 - Develop a graphical user interface
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11. Conclusion

This project successfully demonstrates a real-time color tracking system. It is simple, efficient, and provides a strong foundation for building advanced computer vision applications.